Lab 1: Requirements Engineering & UML Use-Case Modelling

Problem Statement 38: Dropshipping Inventory & Order Orchestrator

SAMARTH KOTIAN
PES1UG24CS406

GITHUB : https://github.com/samarthkotian/software_engineering_lab

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall parse inbound Shopify/WooCommerce order webhooks, verify supplier API inventory, and auto-dispatch supplier purchase orders.	High	Pass: Supplier PO triggered with customer delivery address. Fail: Order placed for out-of-stock supplier SKU.	Automates the core order-to-fulfillment pipeline and prevents orders being accepted for items suppliers cannot fulfill.
FR-002	Functional	The system shall split a multi-vendor customer order into separate sub-orders, one per supplier, when the order's line items are sourced from more than one supplier.	High	Pass: An order with items from 3 different suppliers generates exactly 3 sub-orders, each containing only that supplier's line items plus the shared delivery address. Fail: Any sub-order is missing line items or contains another supplier's items.	Suppliers ship independently, so orders must be decomposed correctly to avoid misrouted or incomplete shipments.
FR-003	Functional	The system shall block auto-dispatch of a purchase order to a supplier whose real-time inventory check returns a quantity below the ordered quantity for any SKU in that order.	High	Pass: An order for 5 units of an SKU with 3 units in supplier stock is blocked pre-dispatch and flagged "Insufficient Stock." Fail: A PO is dispatched despite a stock shortfall.	Prevents committing to purchase orders the supplier cannot fulfill, protecting customer trust and store reputation.
FR-004	Functional	The system shall poll each active supplier's shipment-tracking API (or ingest supplier tracking webhooks) at least once every 4 hours and update the corresponding sub-order's tracking number and carrier status.	Medium	Pass: A tracking number issued by a supplier appears against the correct sub-order within 4 hours. Fail: A sub-order's tracking field remains blank more than 4 hours after the supplier issued a tracking number.	Keeps store owners and customers informed of shipment progress without manual follow-up.
FR-005	Functional	The system shall notify the E-Store Owner, via dashboard alert and email, whenever a purchase-order dispatch to a supplier fails after 2 retry attempts.	Medium	Pass: After 2 consecutive failed dispatch attempts, a dashboard alert appears and an email is sent within 5 minutes.	Ensures exceptions that need human intervention are surfaced promptly instead of failing silently.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
				Fail: No alert is raised after 2 failed attempts, or it is delayed beyond 5 minutes.	
NFR-001	Nonfunctional	Inbound webhook processing must acknowledge receipt with HTTP 200 within 200 ms to prevent provider webhook retry storms.	High	Pass: Benchmarking tests confirm target latency and security standards are met under simulated peak load.	Storefront platforms retry unacknowledged webhooks aggressively; slow responses risk duplicate order ingestion and platform-side rate limiting.
NFR-002	Nonfunctional	The system shall maintain at least 99.5% monthly uptime for the order-ingestion and supplier-dispatch pipeline.	High	Pass: Monthly uptime logs show downtime for the ingestion/dispatch pipeline within the 99% budget Fail: Downtime exceeds this threshold in any calendar month.	Order ingestion and PO dispatch sit on the critical path; extended downtime directly causes lost or delayed sales.

Actors:

1. Storefront Platform
2. Supplier Partner
3. E-Store Owner

Use Cases:

1.UC-01: Process Incoming Order

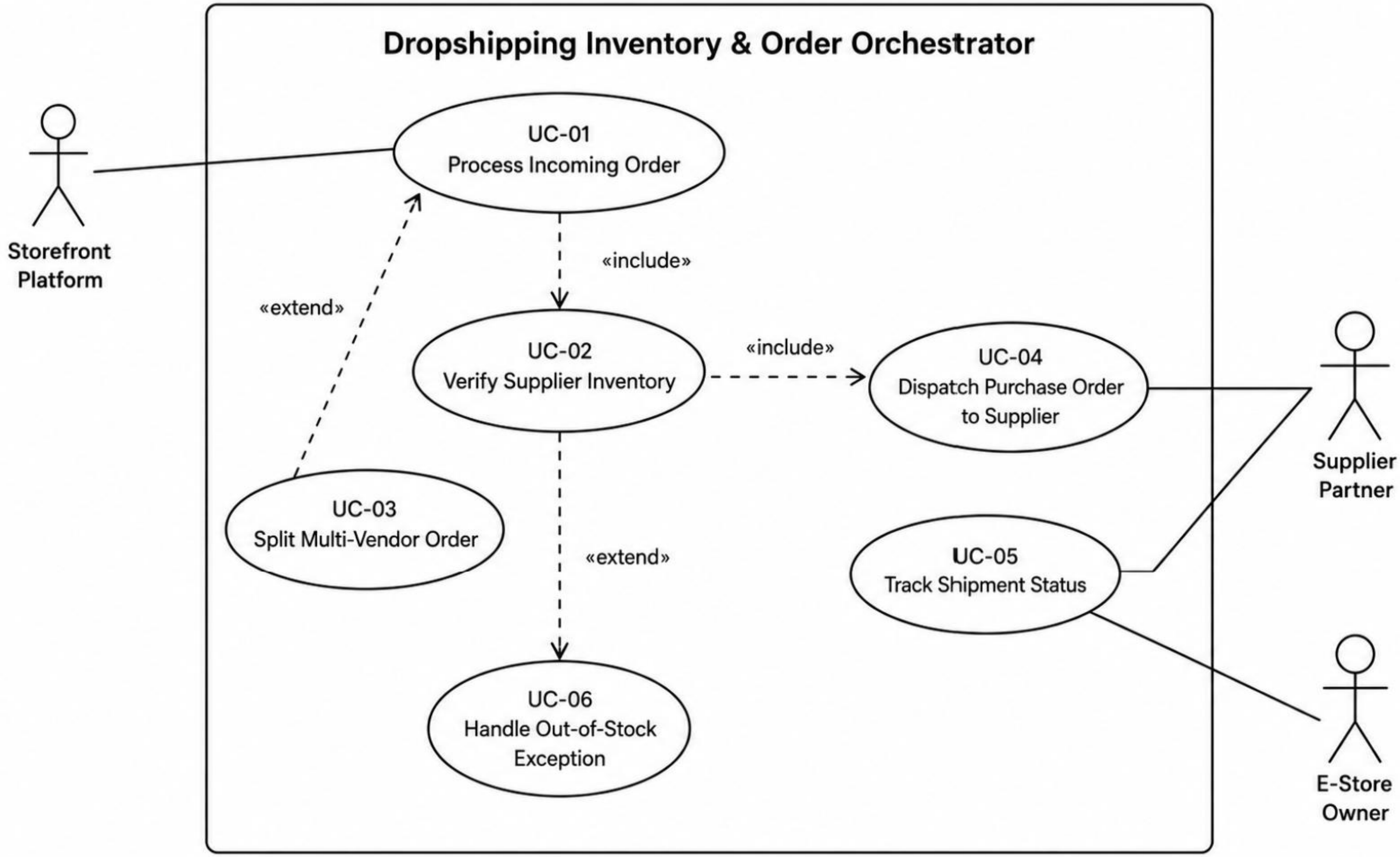
2.UC-02: Verify Supplier Inventory

3.UC-03: Split Multi-Vendor Order

4.UC-04: Dispatch Purchase Order to Supplier

5.UC-05: Track Shipment Status

6.UC-06: Handle Out-of-Stock Exception



Dropshipping Inventory & Order Orchestrator
Storefront Platform
UC-01
Process Incoming Order
«include»
«extend»
UC-02
Verify Supplier Inventory
«include»
UC-04
Dispatch Purchase Order to Supplier
Supplier Partner
UC-03
Split Multi-Vendor Order
«extend»
UC-05
Track Shipment Status
UC-06
Handle Out-of-Stock Exception
E-Store Owner

Use Case: UC-01 – Process Incoming Order

Actors: Storefront Platform ,
Supplier Partner,
E-Store Owner.

Preconditions

- The Storefront Platform's order webhook endpoint is registered and verified in the system.
- At least one Supplier Partner account is active with valid inventory-API credentials.
- Every catalog SKU is mapped to exactly one Supplier Partner.

Postconditions

- Success: A purchase order has been dispatched to each Supplier Partner represented in the order, and the order status is set to “Dispatched to Suppliers”, visible on the E-Store Owner's dashboard.
- Failure/partial: Any sub-order with insufficient stock is flagged “Insufficient Stock” and excluded from dispatch until resolved; unaffected sub-orders still dispatch normally.

Main Success Scenario :

1. Storefront Platform sends an order-created webhook containing customer details, line items, and delivery address to the system's webhook endpoint.
2. System acknowledges the webhook with HTTP 200 within 200 ms and queues the order for processing (NFR-001).
3. System parses the payload and identifies the Supplier Partner associated with each line item's SKU.
4. System queries each identified Supplier Partner's inventory API in real time to verify stock availability for the requested quantities.
5. System confirms sufficient stock is available for every line item.
6. System splits the order into one sub-order per supplier, each carrying the shared delivery address (FR-002).
7. System auto-generates and dispatches a purchase order to each Supplier Partner's API (FR-001).
8. Each Supplier Partner acknowledges receipt of its purchase order.
9. System updates the order status to “Dispatched to Supplier(s)” and logs a confirmation entry visible to the E-Store Owner.
10. Use case ends successfully.

Alternate Flow : Insufficient Supplier Stock

1.System detects that the available quantity for one or more SKUs is less than the ordered quantity (FR-003).
2.System blocks purchase-order dispatch for the affected supplier's sub-order and flags it “Insufficient Stock.”
3.System notifies the E-Store Owner via dashboard alert and email so they can cancel, back-order, or manually source the item.
4.Once the E-Store Owner resolves the flag, the system either resumes dispatch for the corrected quantity or cancels that sub-order; unaffected sub-orders from other suppliers continue through the main flow unaffected.